

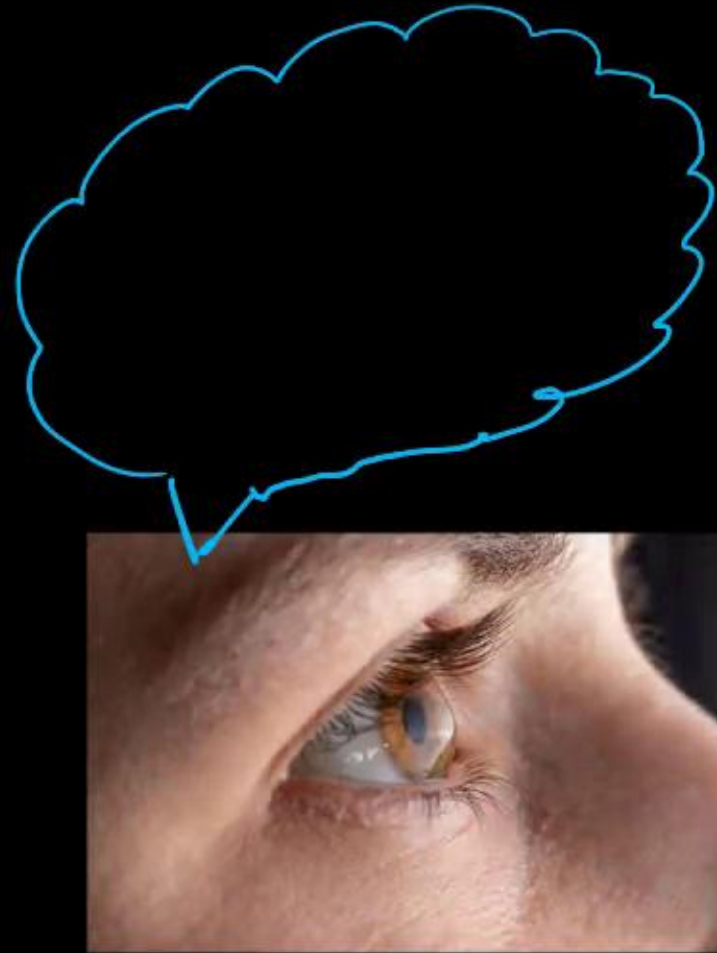
CHAPTER 1 : LIGHT REFLECTION AND REFRACTION

Learning Objectives :

- Introduction
- Reflection And Laws of reflection
- Spherical Mirrors
- Image formation by Spherical Mirrors
- Ray Diagrams
- Uses of Concave and Convex Mirrors
- Sign Convention
- Mirror Formula and Magnification
- Refraction of Light
- Refraction through a Rectangular Glass Slab
- Laws of Refraction
- The Refractive Index
- Refraction by Spherical Lenses
- Image Formation by Lenses & Ray Diagrams
- Lens Formula & Magnification
- Power of a Lens

Introduction

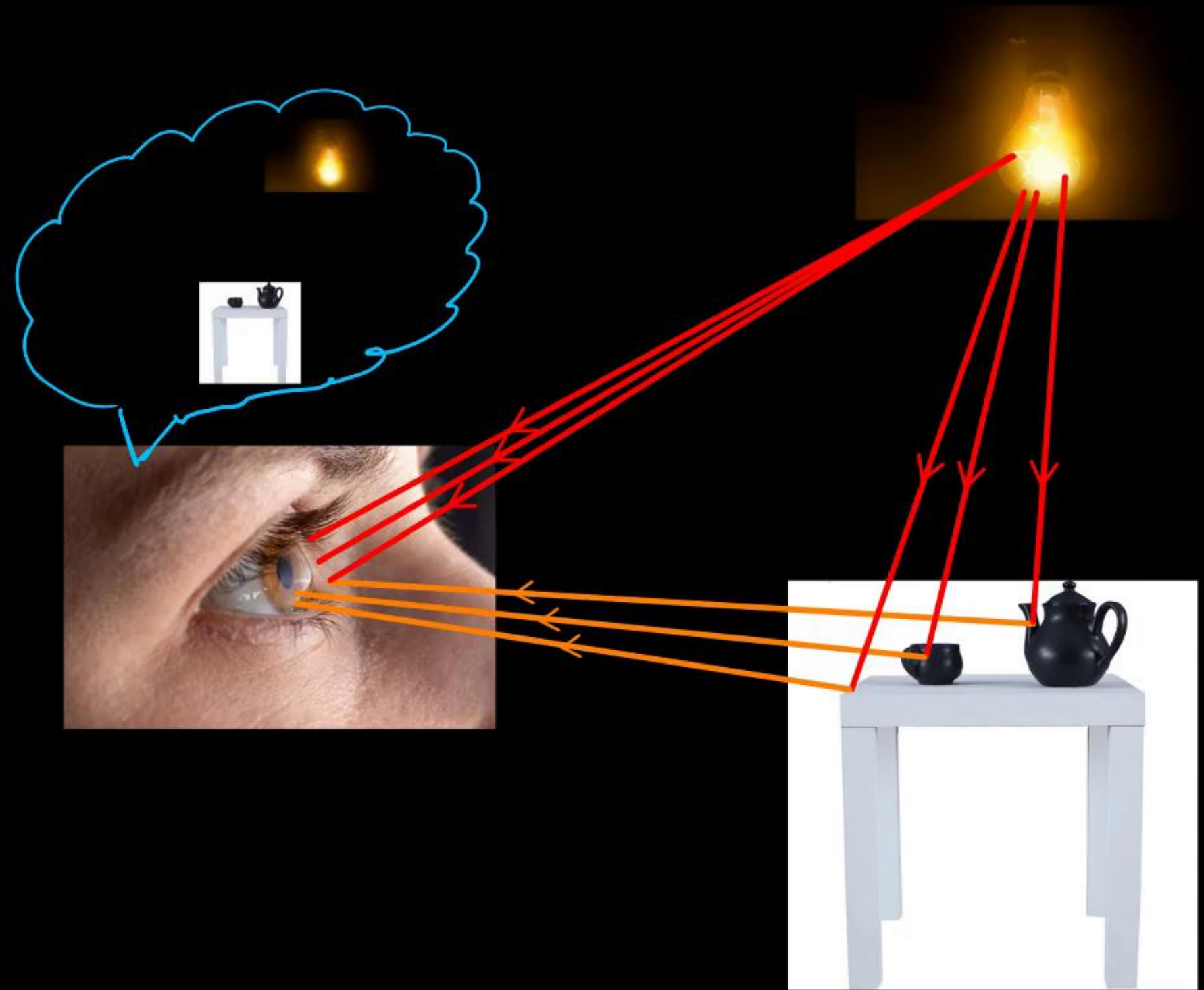
What makes things visible?



Introduction

What makes things visible?

An object emits or reflects light that falls on it. This emitted or reflected light, when received by our eyes, enables us to see things.



Spherical Mirrors

- **Spherical Mirrors** : Mirrors, whose reflecting surface can be considered to form a part of the surface of a sphere. The spherical mirror is of two types:
- (i) **Concave mirror** : A spherical mirror, whose reflecting surface is curved inwards, is called a concave mirror
- (ii) **Convex mirror** : A spherical mirror, whose reflecting surface is curved outwards, is called a convex mirror



Concave mirror

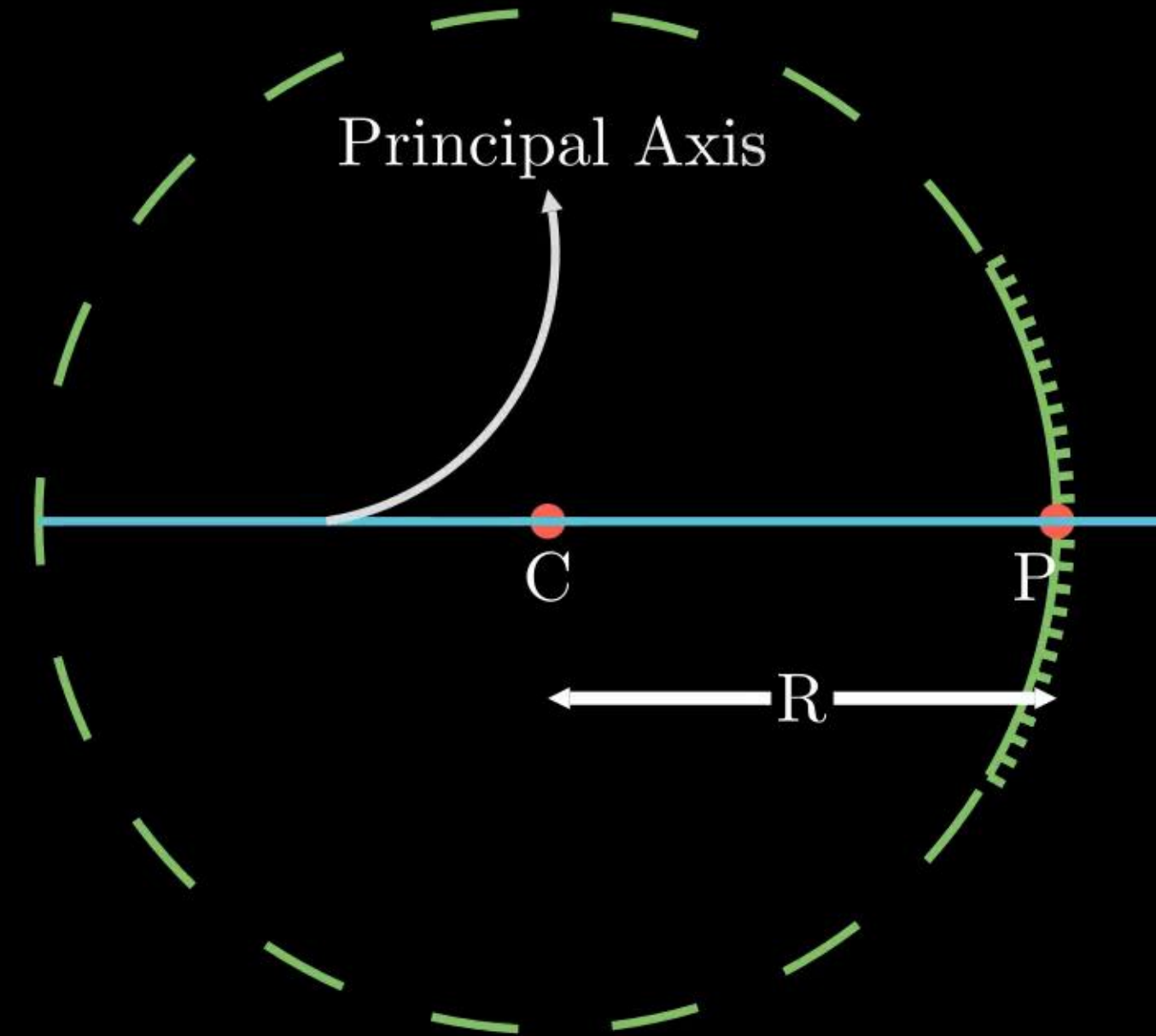


Convex mirror



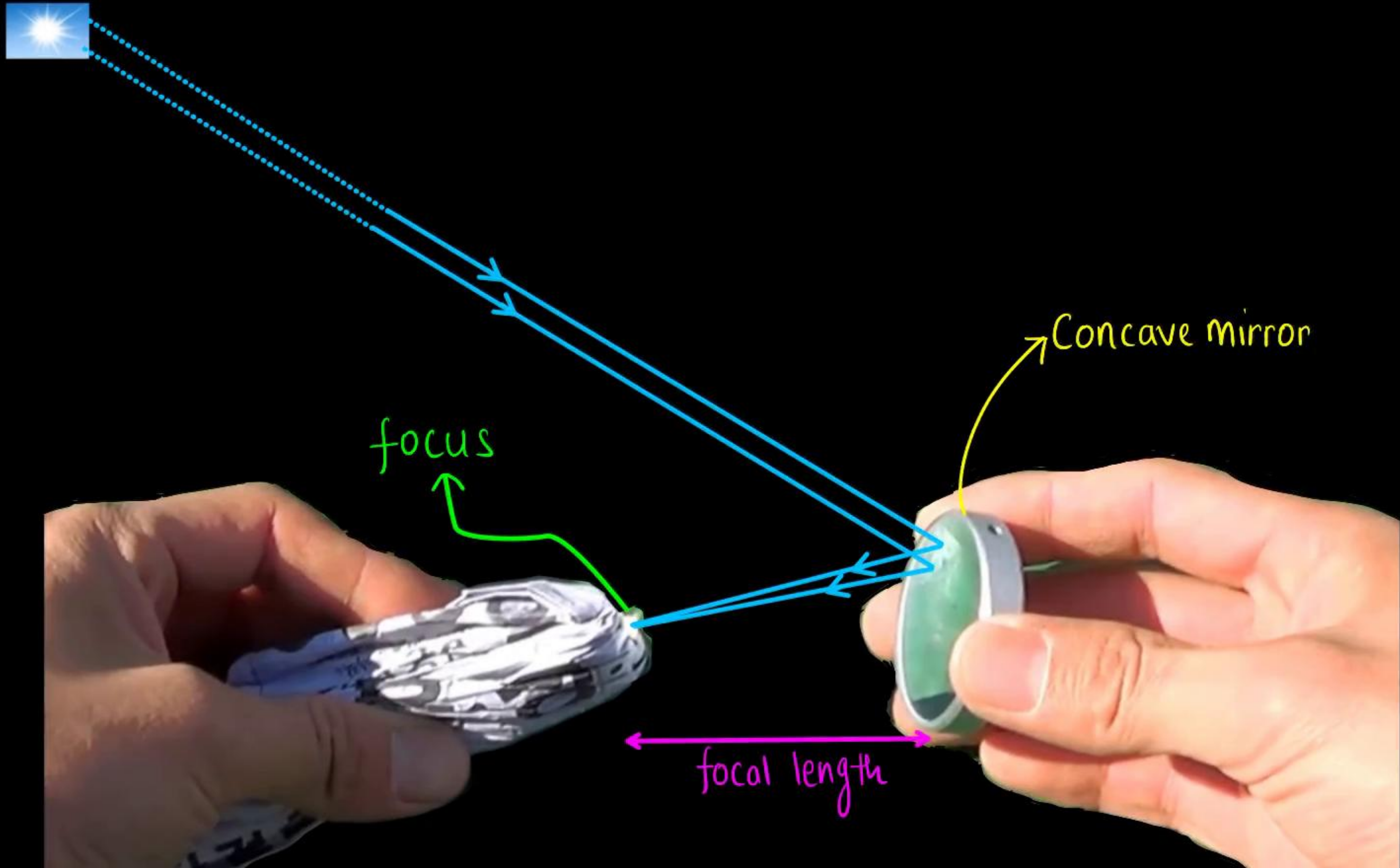
Some Definitions Related to Spherical Mirror:

- **Pole (P)**: The centre of the reflecting surface of a spherical mirror is called the pole. It lies on the surface of the mirror.
- **Centre of curvature (C)**: The centre of the sphere of which the mirror is a part is called the centre of curvature. Note that the centre of curvature is not a part of the mirror. It lies outside its reflecting surface.
- **Radius of curvature (R)** : The radius of the sphere of which the mirror is a part is called the radius of curvature. Or the distance (PC) between pole and centre of curvature.
- **Principal axis** : An imaginary straight line passing through the pole and the centre of curvature of the mirror is called the principal axis.



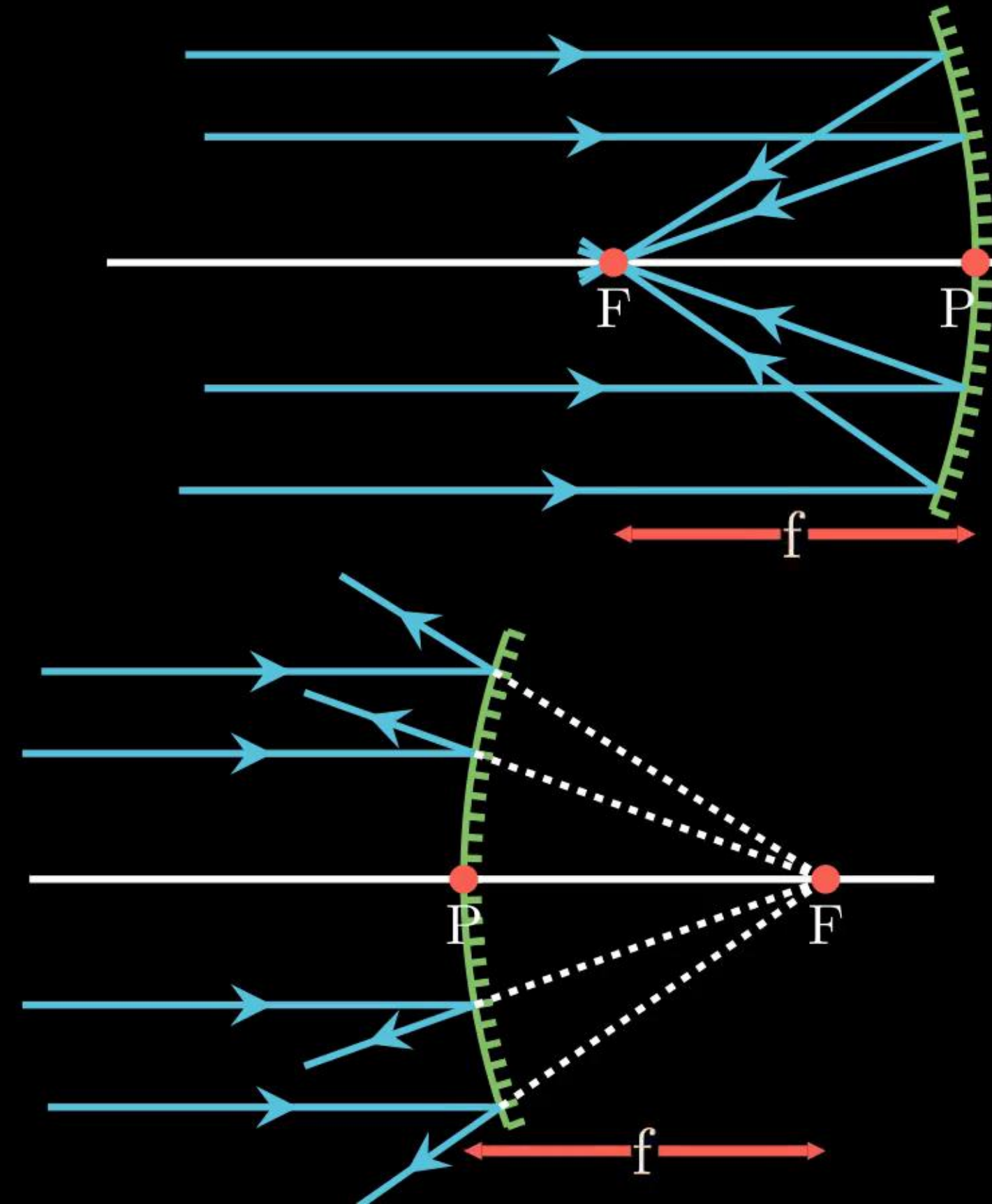
Some Definitions Related to Spherical Mirror:

Activity :



Some Definitions Related to Spherical Mirror:

- **Principal Focus (F) of Concave Mirror:** When light rays parallel to principal axis incident on a concave mirror, after reflection they all meet at a point on the principal axis. This point is called principal focus (F) of concave mirror.
- **Principal Focus (F) of Convex Mirror:** When light rays parallel to principal axis incident on a convex mirror, after reflection they all appear to come from a point on the principal axis. This point is called principal focus (F) of convex mirror.
- **Focal Length (f) :** The distance between the pole (P) and the principal focus (F) is called the focal length



Some Definitions Related to Spherical Mirror:

- **Aperture** :The reflecting surface has a circular outline. The diameter of the reflecting surface of spherical mirror is called its aperture
- **Relation between the radius of curvature (R) and Focal Length(f):**For spherical mirrors of small apertures, the radius of curvature is found to be equal to twice the focal length.

$$R = 2 \times f$$

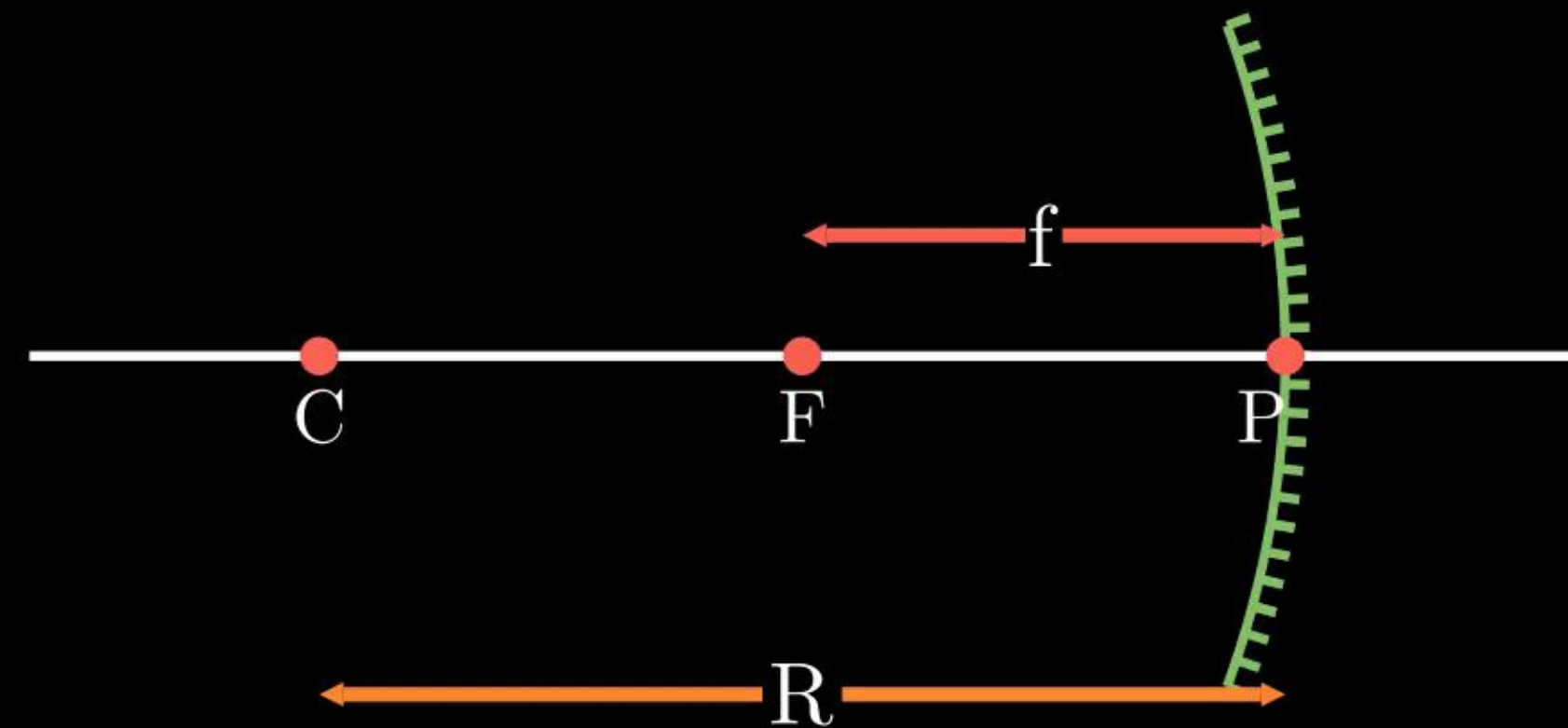


Image formation by Spherical Mirrors

- **Image:** If light rays coming from a point after reflection meet at another point or appear to meet at another point, then second point is called image of the first point. There are two types of image, i.e.-
- **(a) Real image:** When the rays of light, after reflection from a mirror, actually meet at a point, then the image formed by these rays is said to be real. Real images can be obtained on a screen.
- **(b) Virtual image:** When the rays of light, after reflection from a mirror, appear to meet at a point, then the image formed by these rays is said to be virtual. Virtual images can't be obtained on a screen.

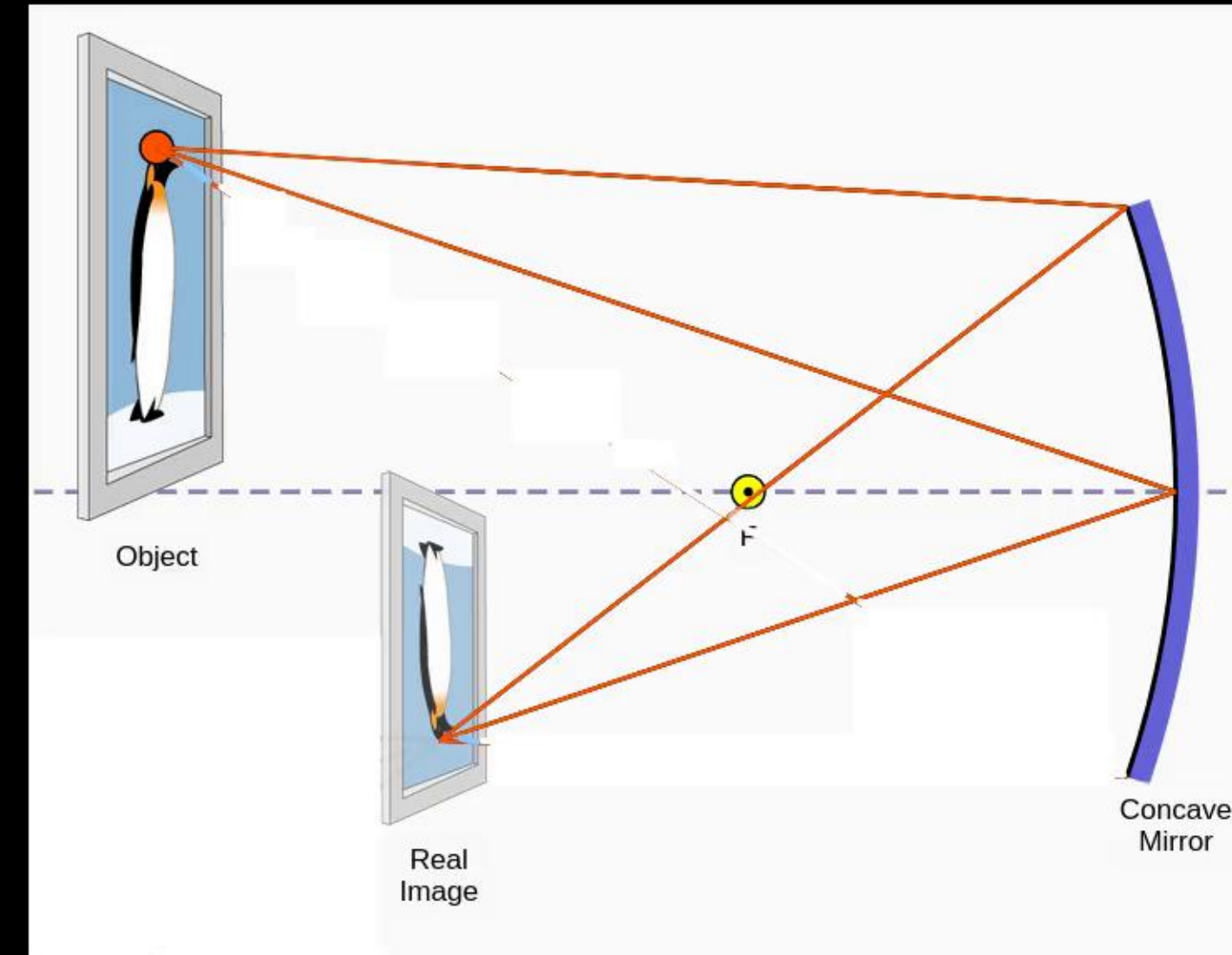
Image formation by Spherical Mirrors

Activity : Image formation by a concave mirror for different positions of the object

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At the focus F	Highly diminished, point-sized	Real and inverted
(2)	Beyond C	Between F and C	Diminished	Real and inverted
(3)	At C	At C	Same size	Real and inverted
(4)	Between C and F	Beyond C	Enlarged	Real and inverted
(5)	At F	At infinity	Highly enlarged	Real and inverted
(6)	Between P and F	Behind the mirror	Enlargede	Virtual and erect

Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

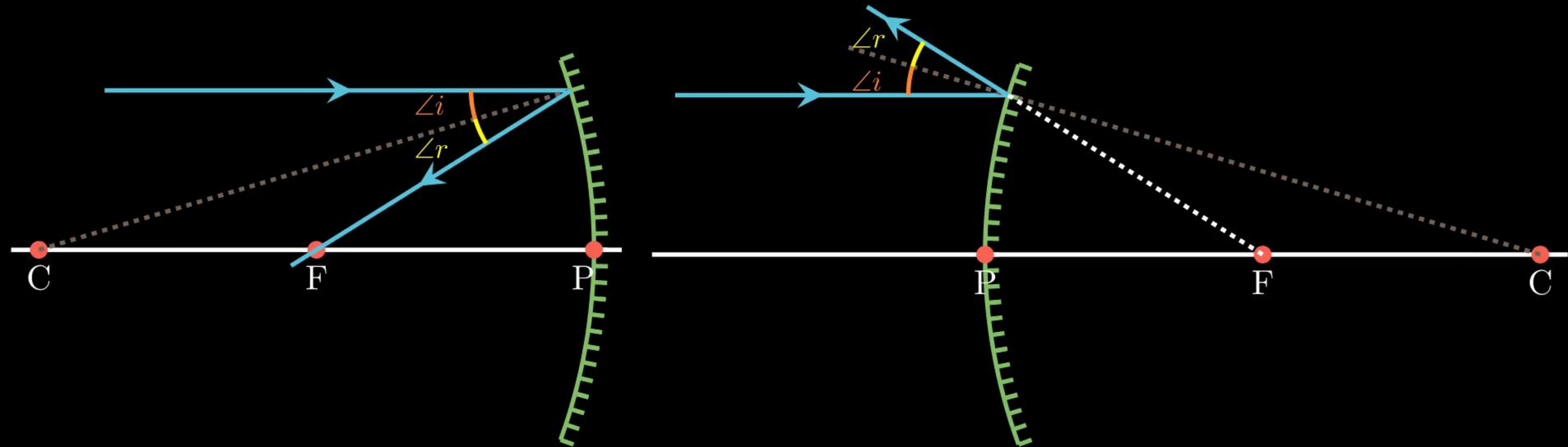
- Consider an extended object, of finite size, placed in front of a spherical mirror.
- Each small portion of the extended object acts like a point source.
- An infinite number of rays originate from each of these points.
- However, it is more convenient to consider only two rays, for the sake of clarity of the ray diagram.
- The intersection of at least two reflected rays give the position of image of the point object.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

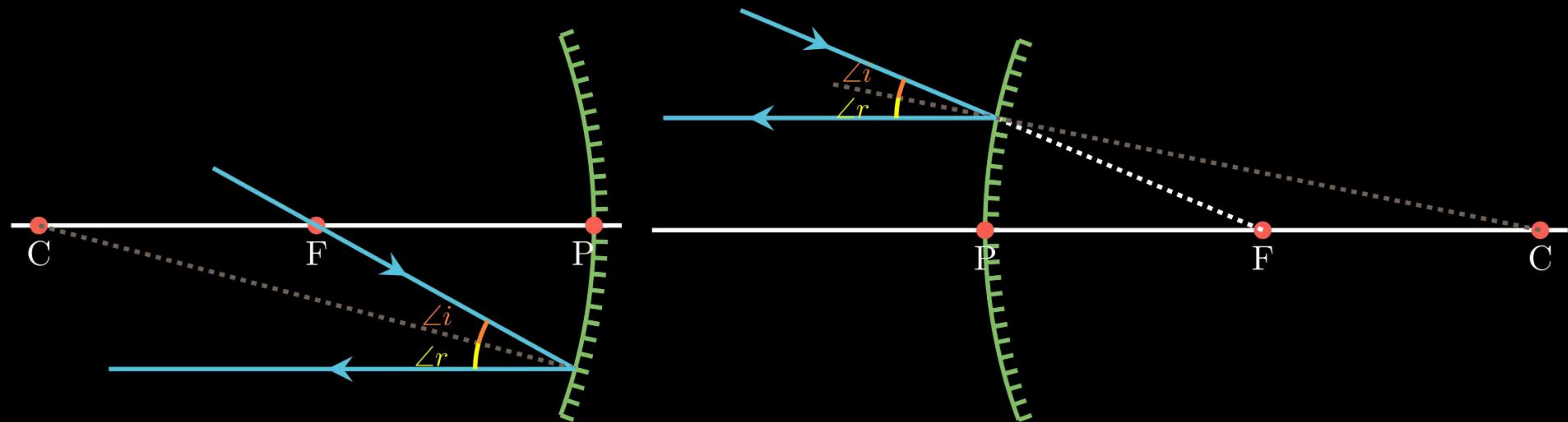
- (i) A ray of light which is parallel to the principal axis of a spherical mirror, after reflection, will pass through the principal focus in case of a concave mirror or appear to diverge from the principal focus in case of a convex mirror.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

- (ii) A ray passing through the principal focus of a concave mirror or a ray which is directed towards the principal focus of a convex mirror, after reflection, will emerge parallel to the principal axis.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

- (iii) A ray passing through the centre of curvature of a concave mirror or directed in the direction of the centre of curvature of a convex mirror, after reflection, is reflected back along the same path.

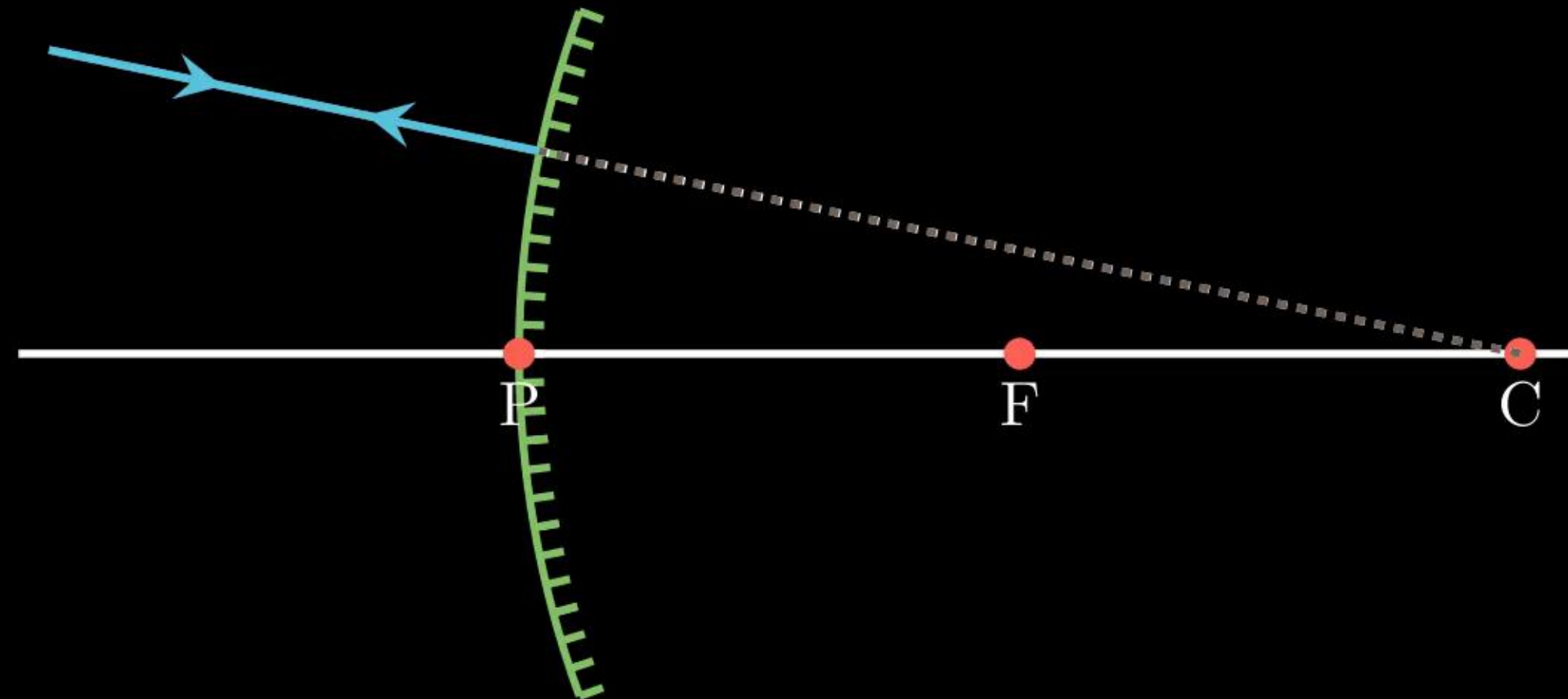
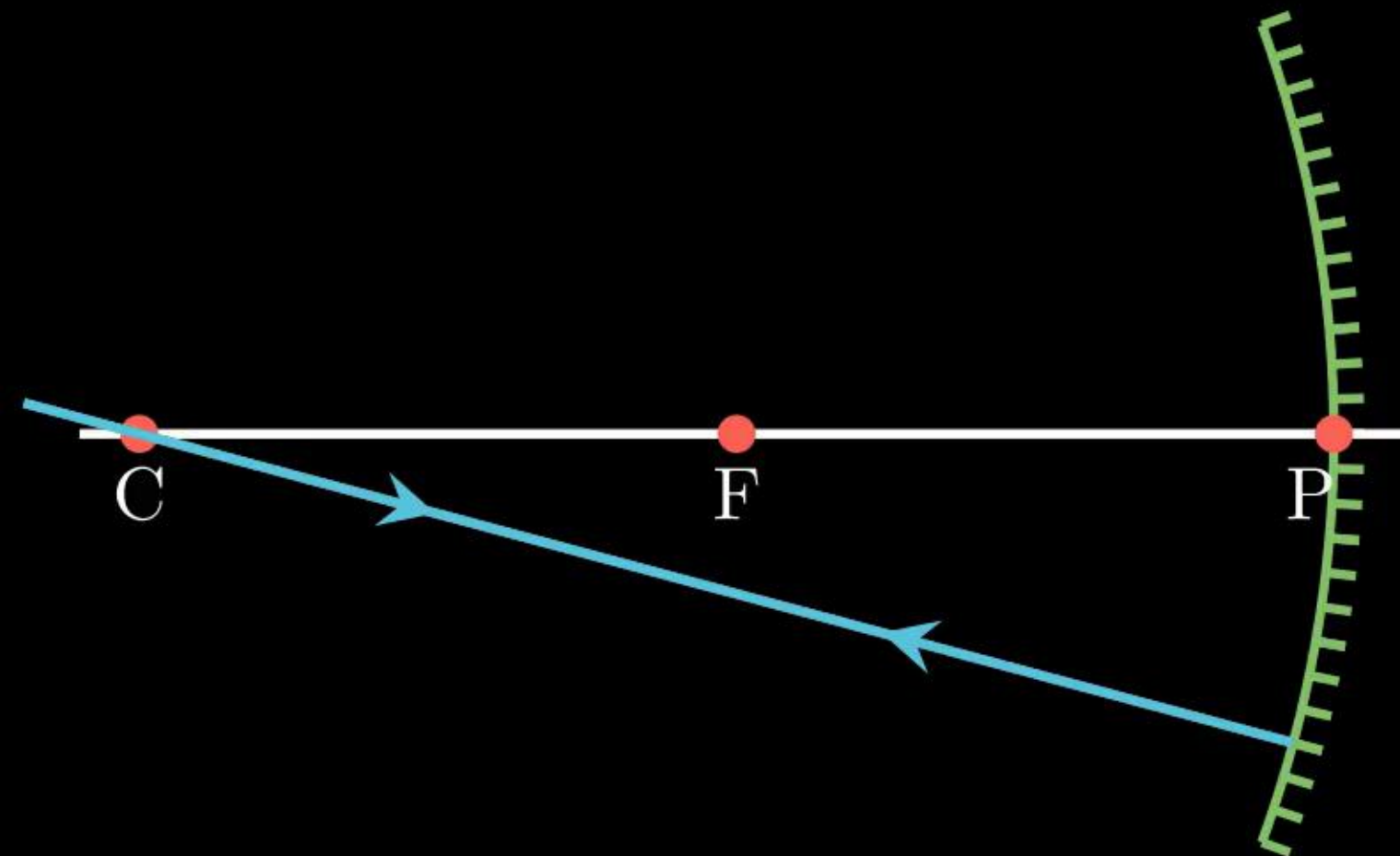
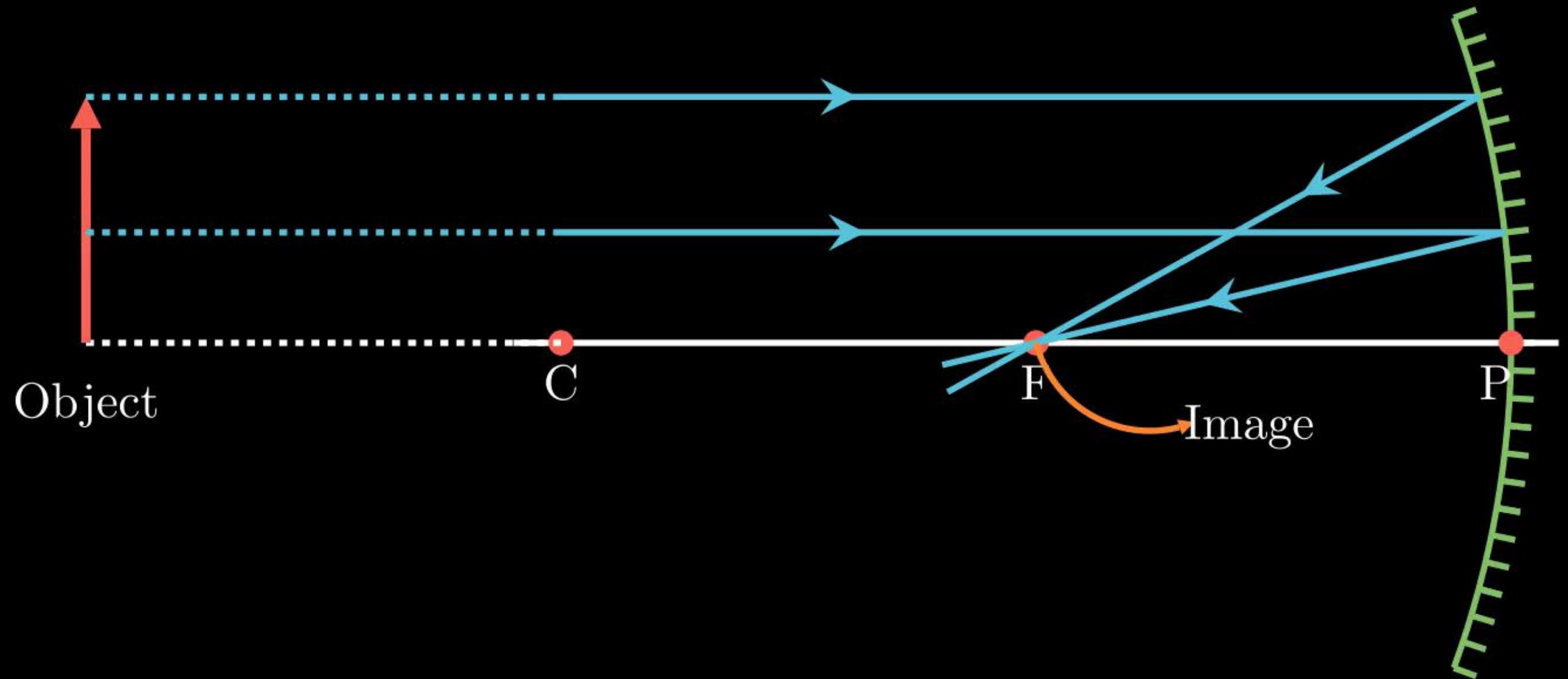


Image formation by Concave Mirror

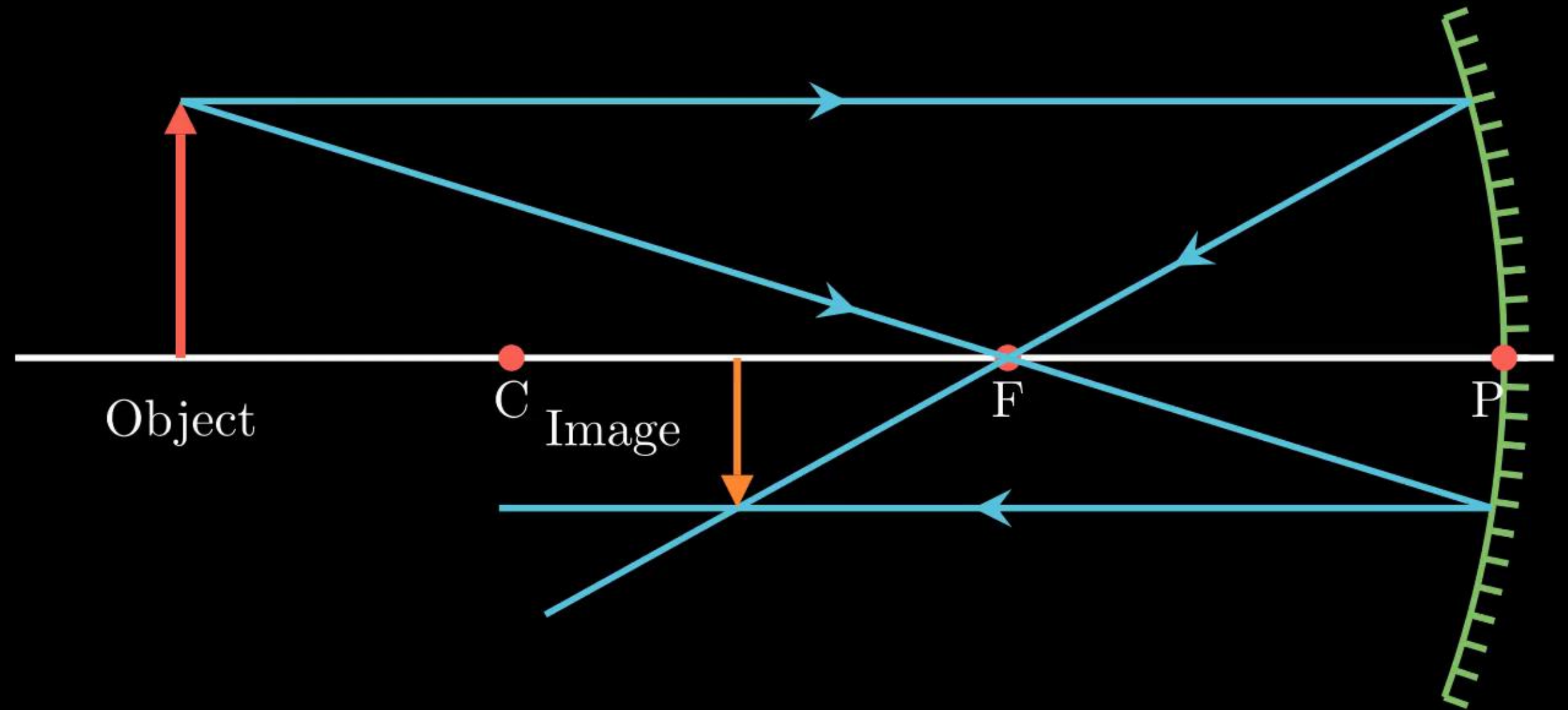
Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus F	Highly diminished, point-sized	Real and inverted

Image formation by Concave Mirror

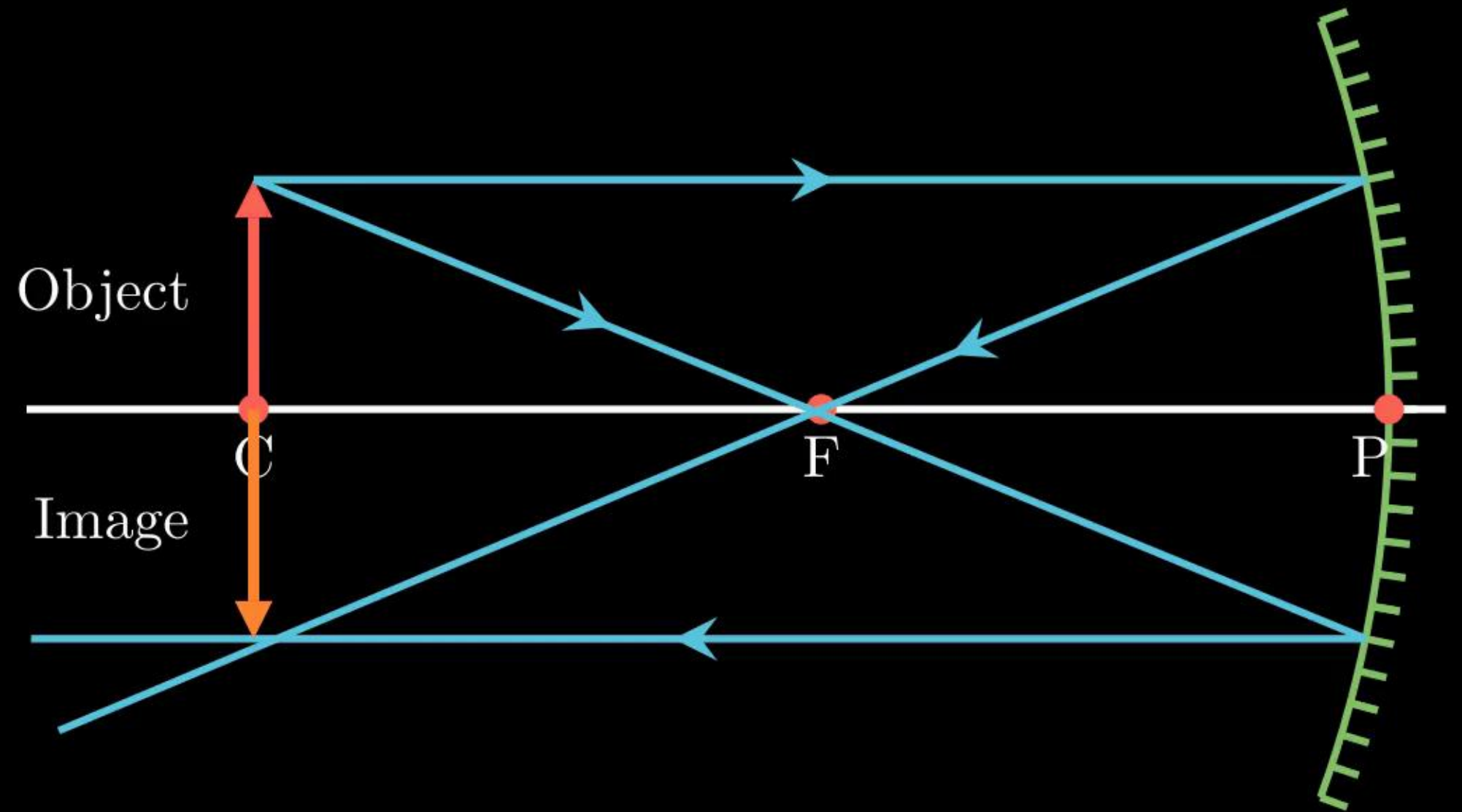
Position of Object (ii): Beyond C



Position of the Image	Size of the Image	Nature of the Image
Between F and C	Diminished	Real and inverted

Image formation by Concave Mirror

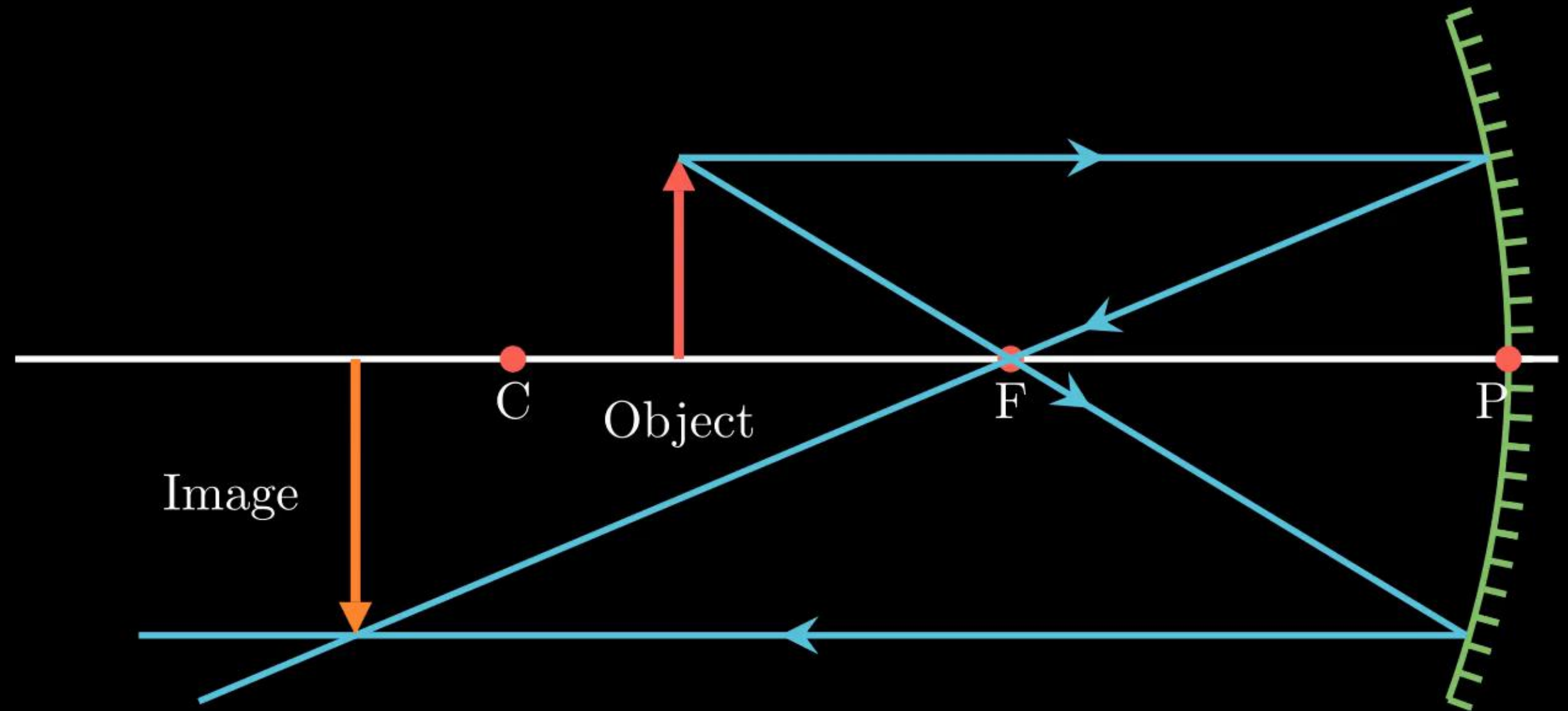
Position of Object (iii): At C



Position of the Image	Size of the Image	Nature of the Image
At C	Same size	Real and inverted

Image formation by Concave Mirror

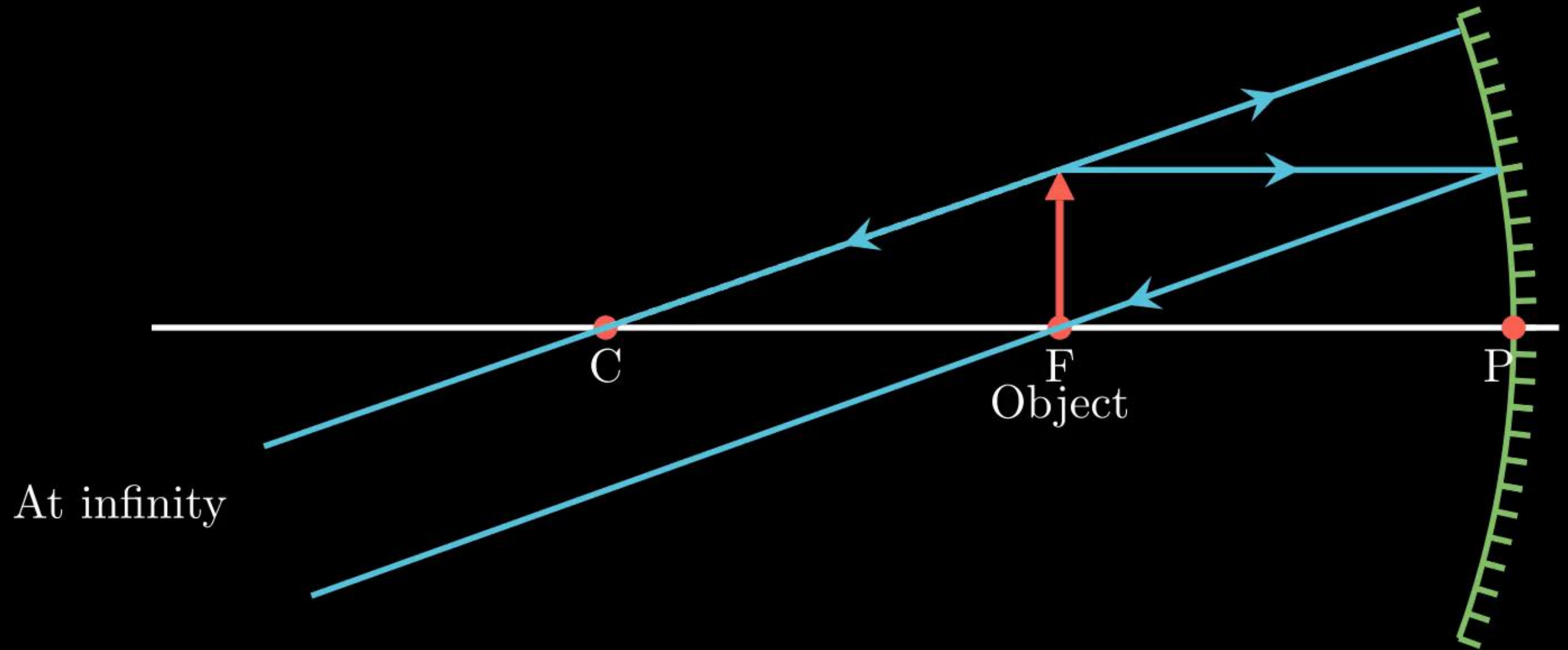
Position of Object (iv): Between C and F



Position of the Image	Size of the Image	Nature of the Image
Beyond C	Enlarged	Real and inverted

Image formation by Concave Mirror

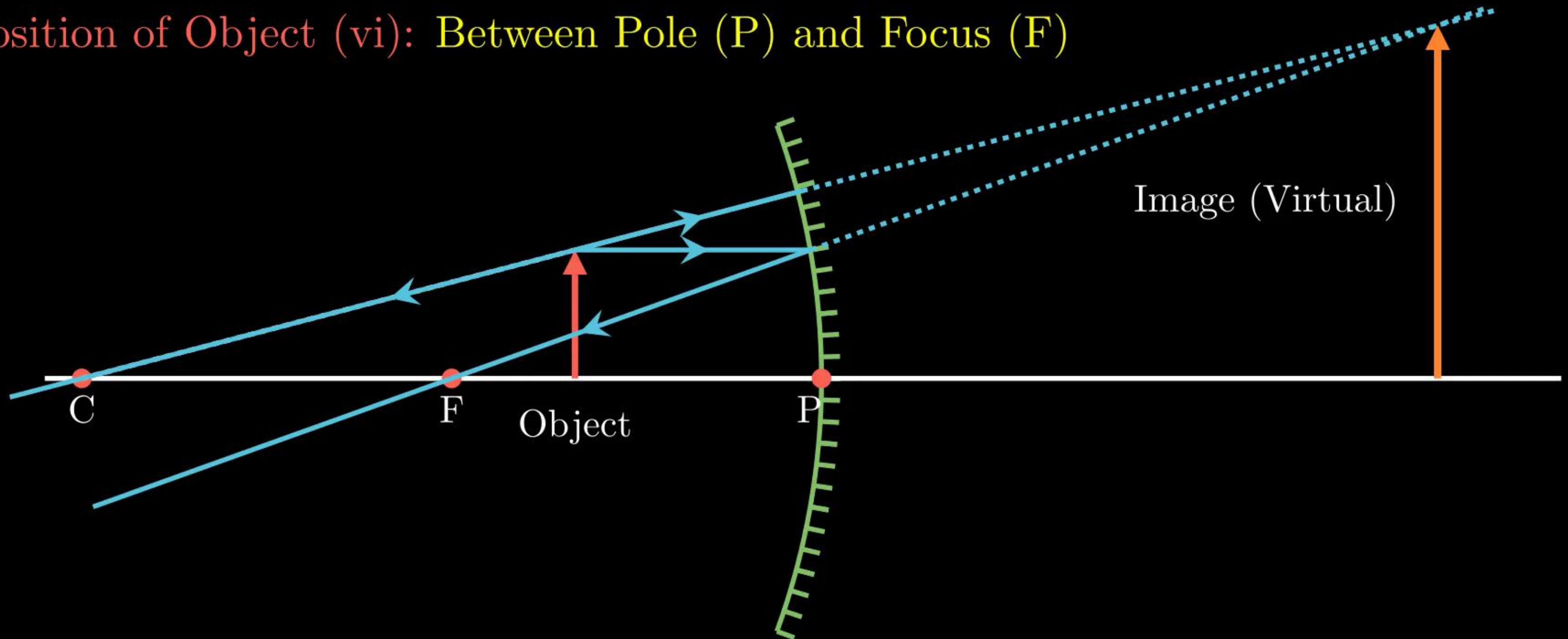
Position of Object (v): At F



Position of the Image	Size of the Image	Nature of the Image
At infinity	Highly enlarged	Real and inverted

Image formation by Concave Mirror

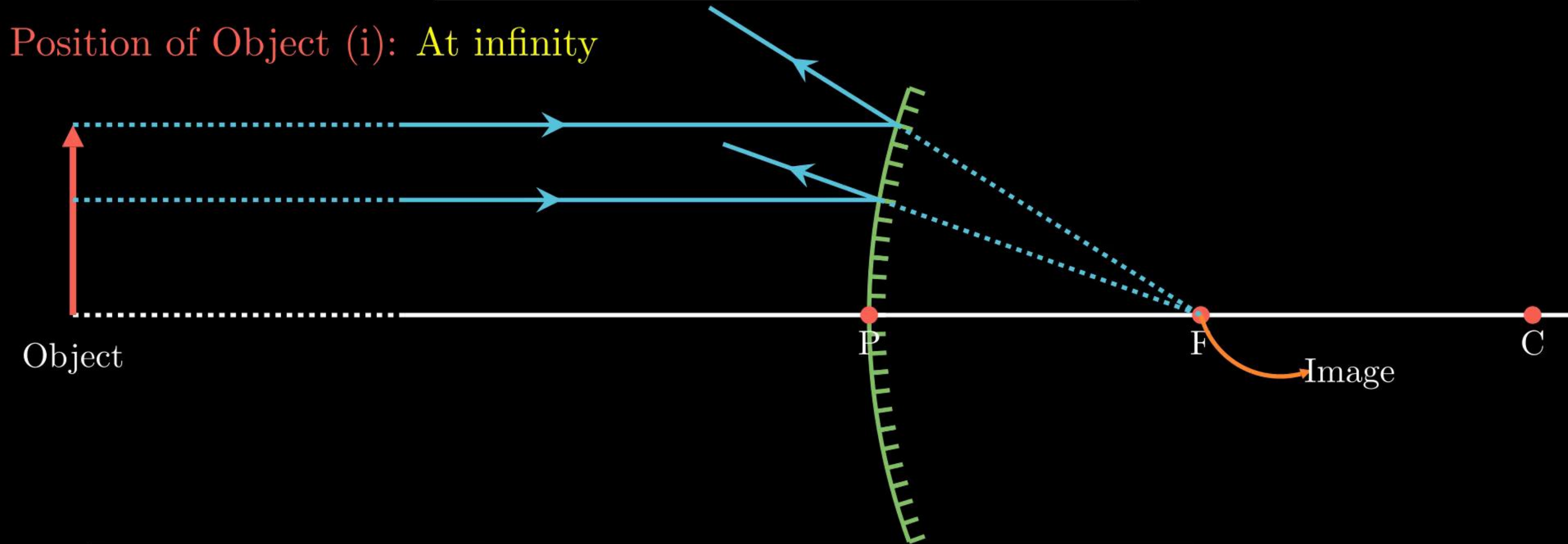
Position of Object (vi): Between Pole (P) and Focus (F)



Position of the Image	Size of the Image	Nature of the Image
Behind the mirror	Enlarged	Virtual and erect

Image formation by Convex Mirror

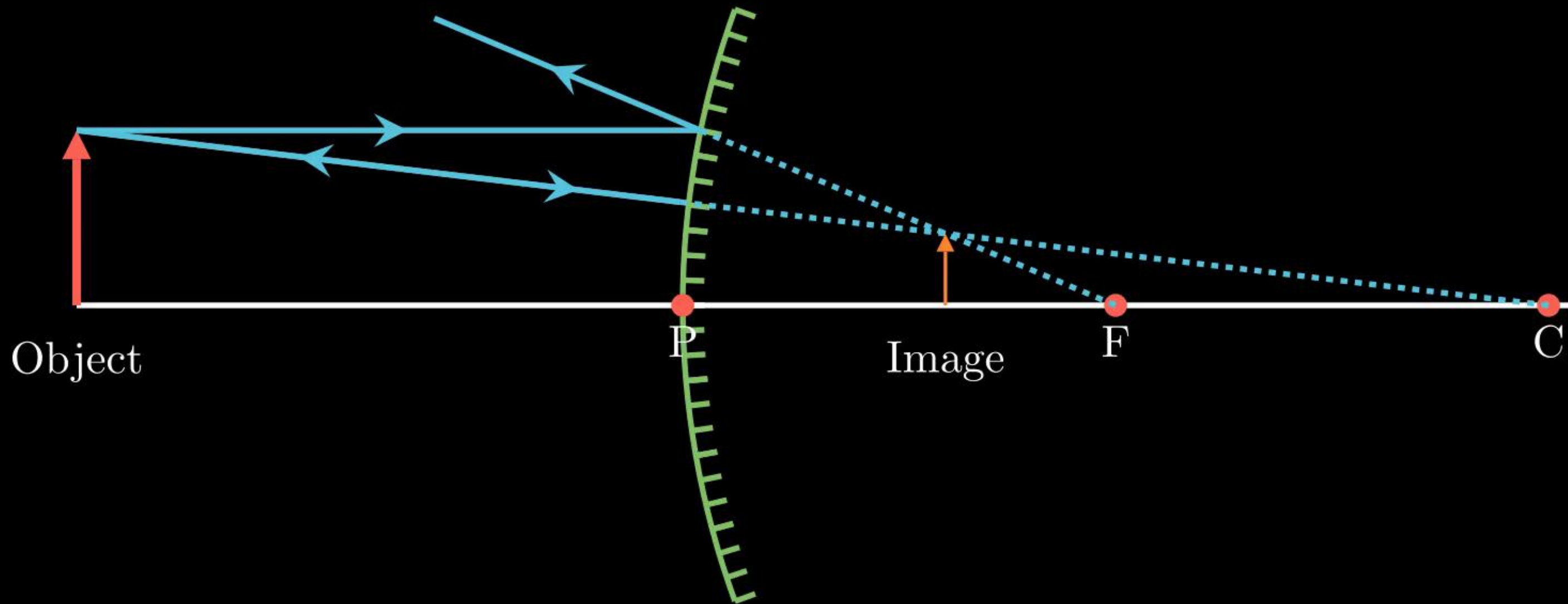
Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect

Image formation by Convex Mirror

Position of Object (ii): Between infinity and the pole P of the mirror

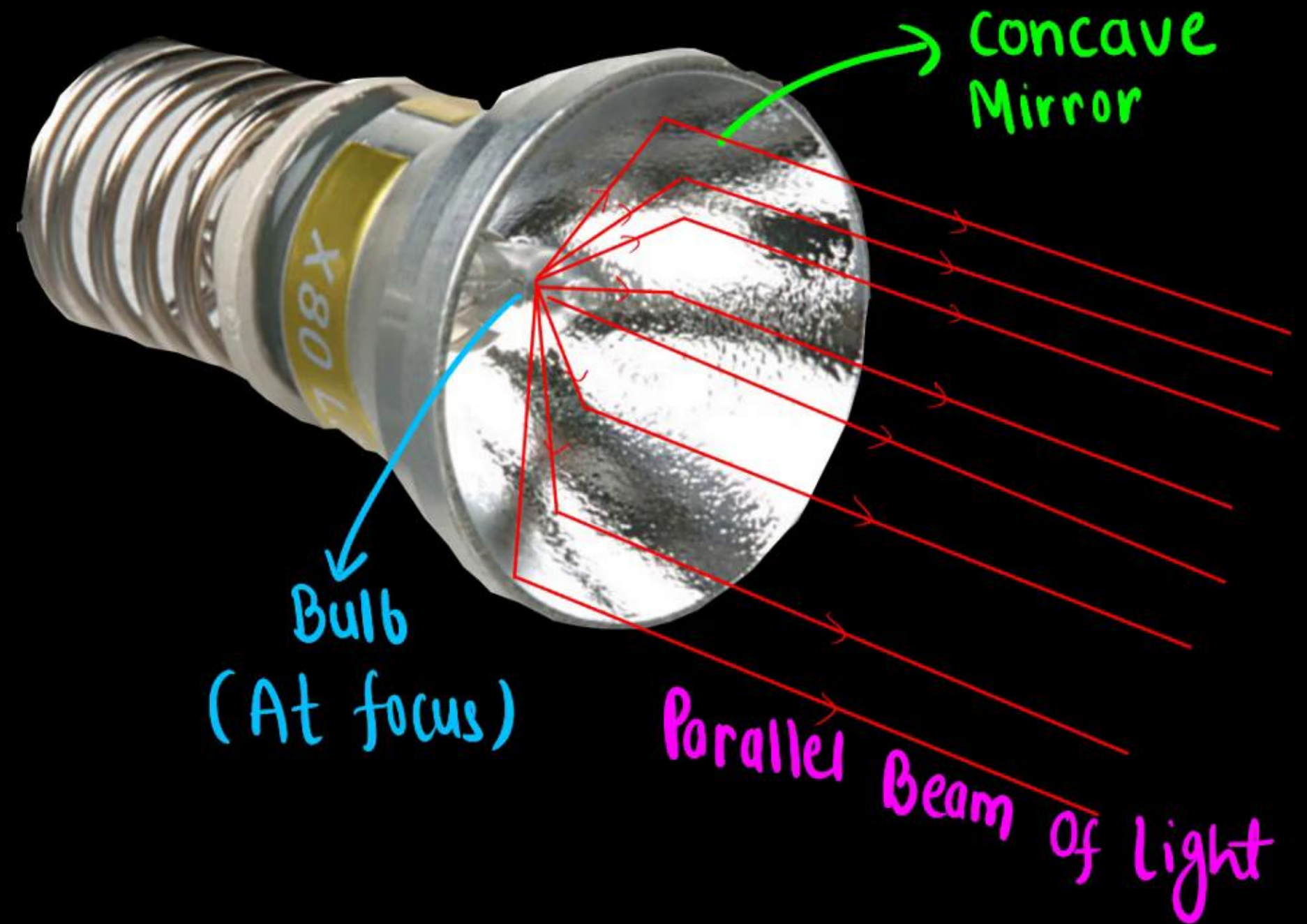


Position of the Image	Size of the Image	Nature of the Image
Between P and F, behind the mirror	Diminished	Virtual and erect

Uses of Concave and Convex Mirrors

Uses of Concave Mirrors :

- (1) Concave mirrors are commonly used in torches, search-lights and vehicles headlights to get powerful parallel beams of light.



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- (3) The dentists use concave mirrors to see large images of the teeth of patients.



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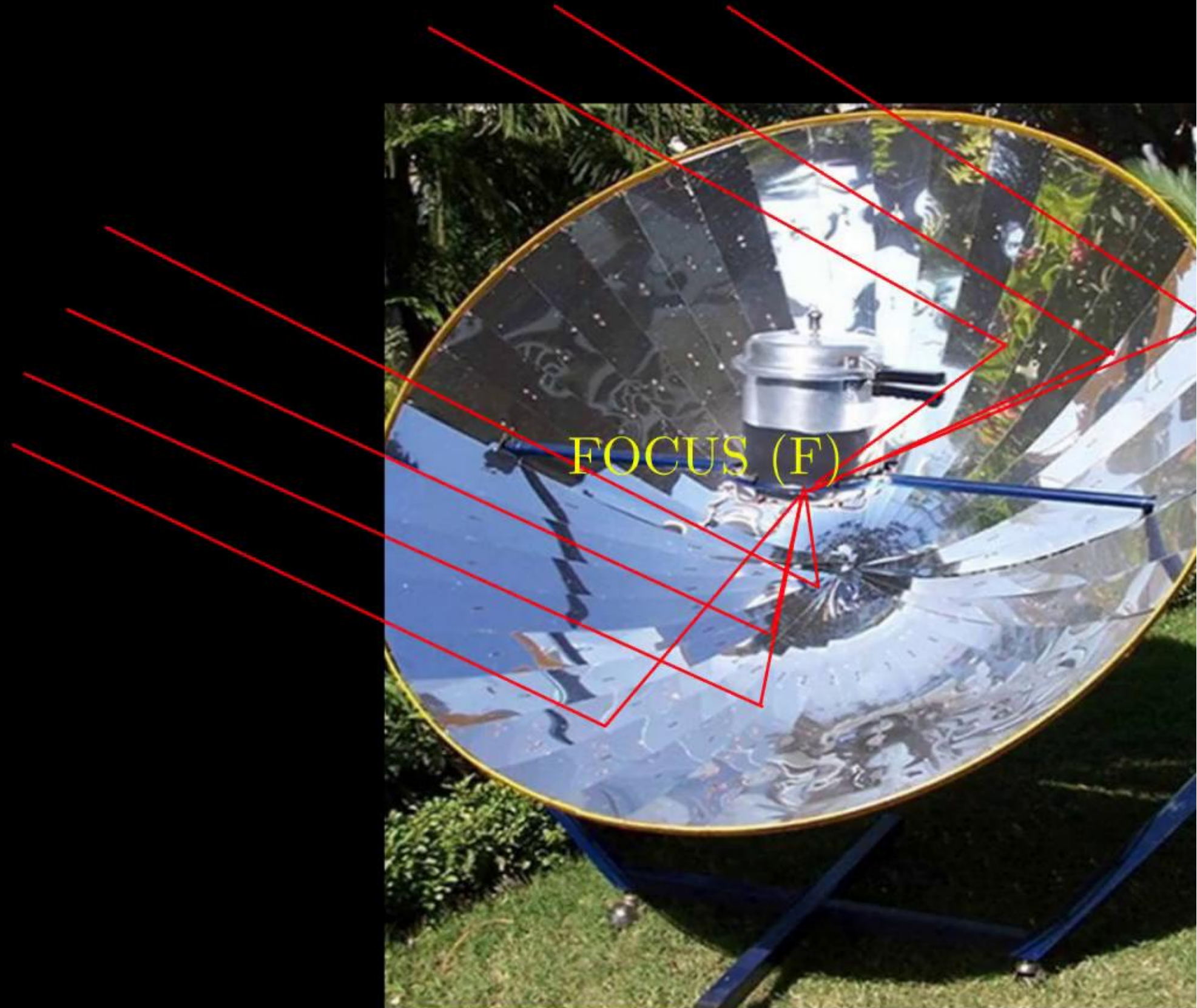
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Uses of Concave and Convex Mirrors

Uses of Convex Mirrors :

- Convex mirrors are commonly used as rear-view (wing) mirrors in vehicles.
- Convex mirrors are preferred because they always give an erect, though diminished, image.
- Also, they have a wider field of view as they are curved outwards. Thus, convex mirrors enable the driver to view much larger area than would be possible with a plane mirror .



Uses of Concave and Convex Mirrors

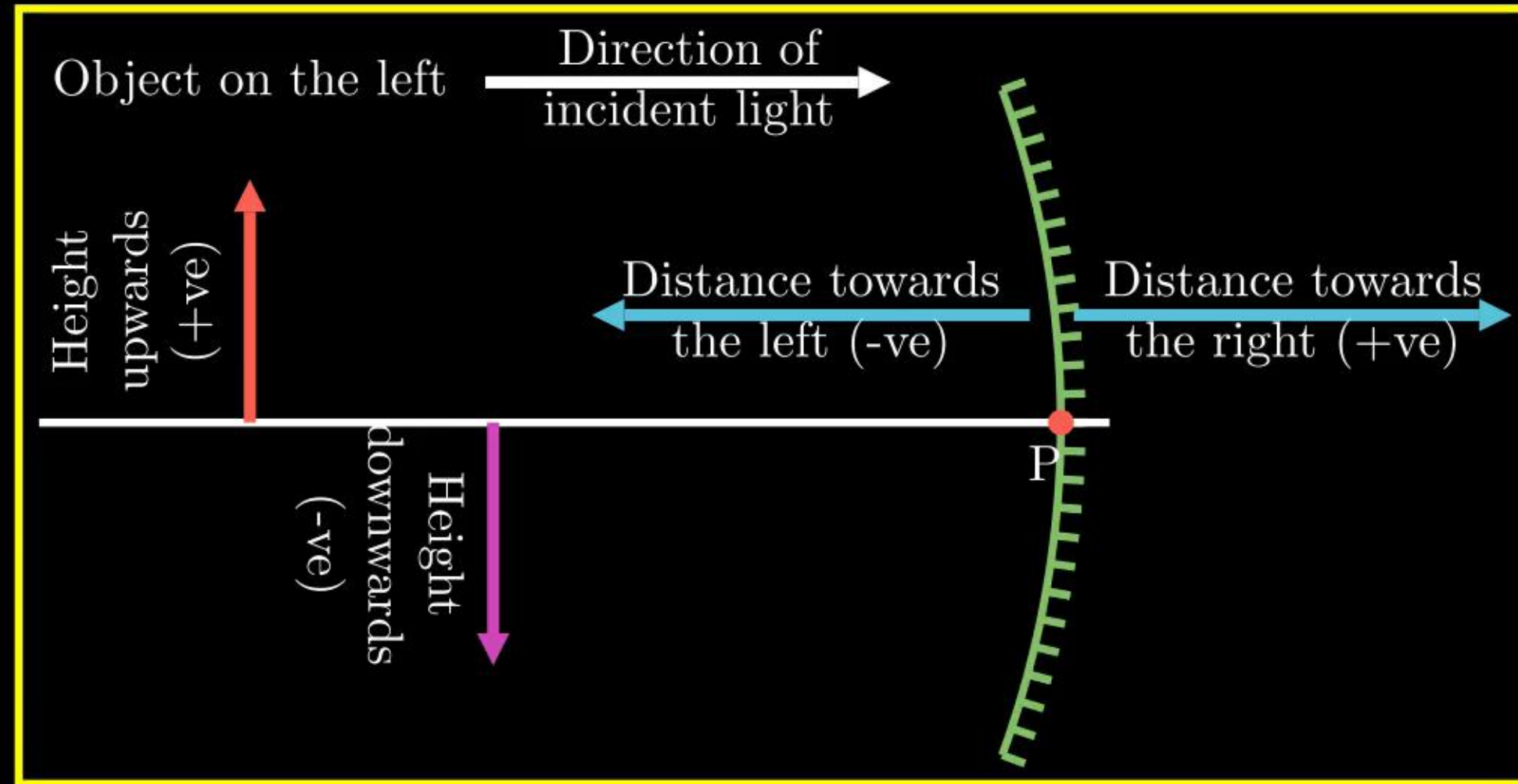
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Sign Conventions For Spherical Mirror

- (i) The object is always placed to the left of the mirror.
- (ii) All distances parallel to the principal axis are measured from the pole of the mirror
- (iii) All the distances measured to the right of the origin (along $+x$ -axis) are taken as positive while those measured to the left of the origin (along $-x$ -axis) are taken as negative.
- (iv) Distances measured perpendicular to and above the principal axis (along $+y$ -axis) are taken as positive.
- (v) Distances measured perpendicular to and below the principal axis (along $-y$ -axis) are taken as negative.

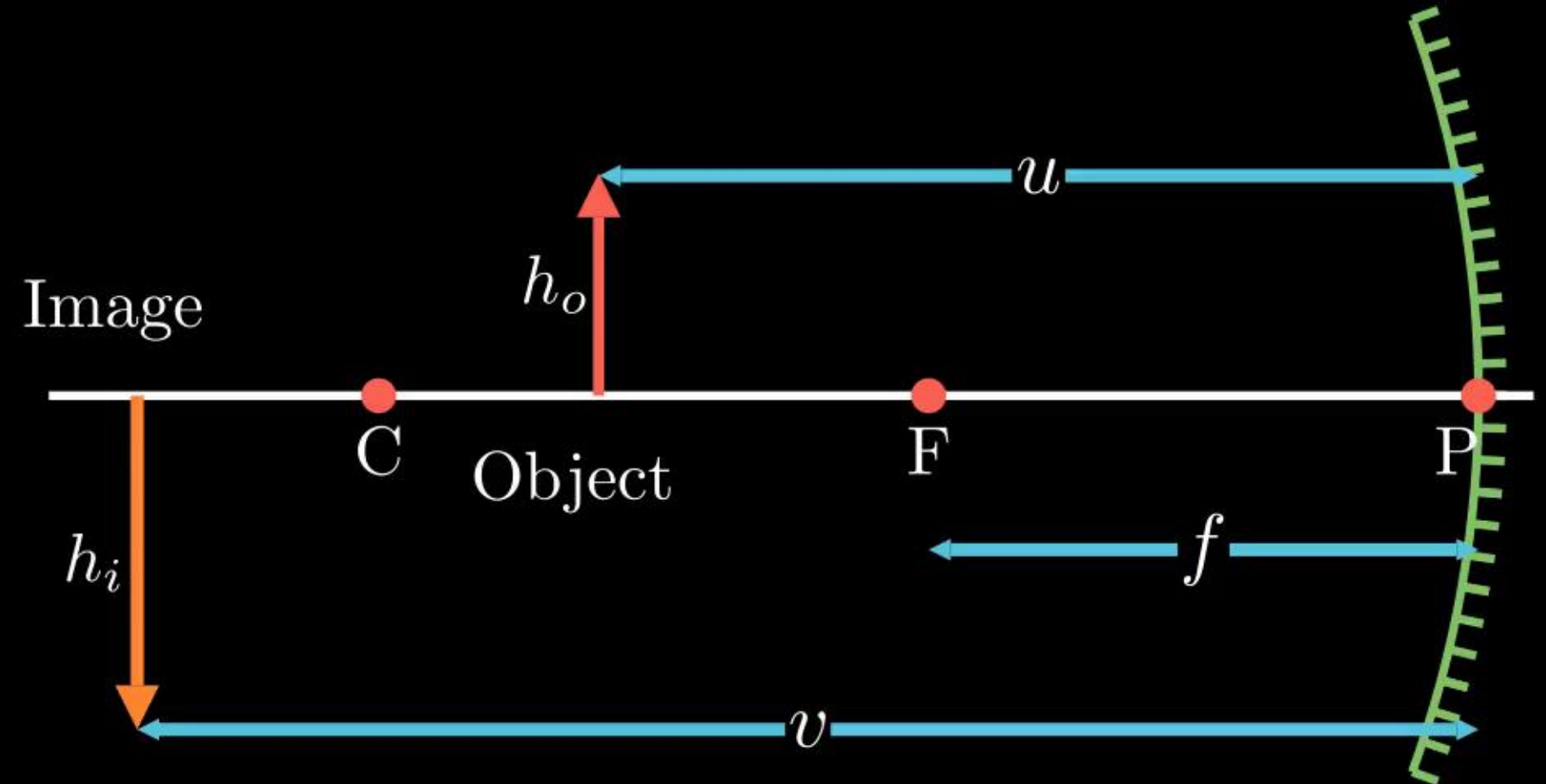


Mirror Formula and Magnification

- **Object Distance (u)** : The distance of the object from the pole.
- **Image Distance (v)** : The distance of the image from the pole.
- **Focal Length (f)** : The distance of the principal focus from the pole.
- **Mirror Formula:** For spherical mirrors, the relationship between u , v and f is given by mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

This formula is valid in all situations for all spherical mirrors for all



- **Object Distance (u)** : Always $(-ve)$
- **Image Distance (v)** : $(+ve)$ if image is virtual, $(-ve)$ if image is real.
- **Focal Length (f)** : $(+ve)$ for convex mirror and $(-ve)$ for concave mirror.

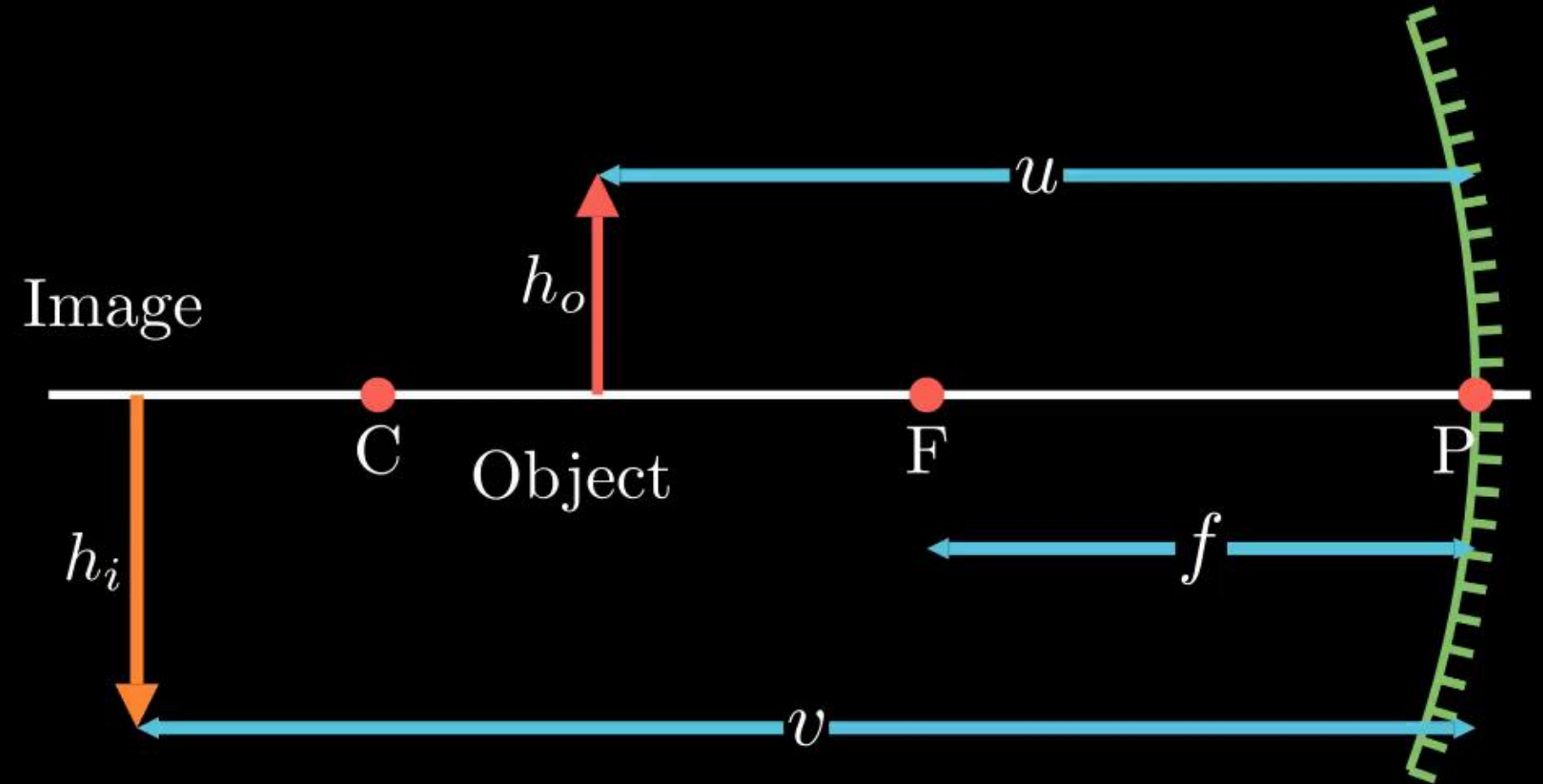
Mirror Formula and Magnification

- (h_o) : Height of the object
- (h_i) : Height of the image.
- **Magnification (m)** : It is the ratio of height of image (h_i) to the height of the object (h_o)

$$m = \frac{h_i}{h_o}$$

- **The magnification m** is also related to the object distance (u) and image distance (v).

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$



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- If $h_i = h_o$, then $m = 1$ i.e., image is equal to object
- If $h_i > h_o$, then $m > 1$ i.e., image is enlarged.
- If $h_i < h_o$, then $m < 1$ i.e., image is diminished.
- If m is +ve, Image is virtual and erect
- If m is -ve, Image is real and inverted
- For Plane Mirror : $m = +1$
- For Convex Mirror : m is +ve and less than 1
- For Concave Mirror : m can be +ve or -ve

Example 1 : Find the focal length of a convex mirror whose radius of curvature is 32 cm.

Solution :

Example 2 : An object, 4.0 cm in size, is placed at 25.0 cm in front of a concave mirror of focal length 15.0 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Find the nature and the size of the image.

Solution :

Example 3 : An object, 4.0 cm in size, is placed at 25.0 cm in front of a concave mirror of focal length 15.0 cm. At what distance from the mirror should a screen be placed in order to obtain a sharp image? Find the nature and the size of the image.

Solution :

Example 4 : A concave mirror produces three times magnified (enlarged) real image of an object placed at 10 cm in front of it. Where is the image located?

Solution :

Example 5 : Identify the mirror used by a dentist to examine a small cavity in a patient's teeth?

(a) Convex mirror

(b) Plane Mirror

(c) Concave Mirror

(d) Any spherical mirror

Solution:

Example 6 : When an object is kept within the focus of a concave mirror, an enlarged image is formed behind the mirror. This image is:

(a) Real

(b) Inverted

(c) Virtual and erect

(d) Virtual and inverted

Solution:

Example 7 : Rays from sun converge at a point 15 cm in front of a concave mirror. Where should an object be placed so that size of its image is equal to that of the object?

(a) 30 cm in front of a mirror

(b) 15 cm in front of a mirror

(c) Between 15 cm and 30 cm in front of a mirror

(d) More than 30 cm in front of a mirror

Solution:

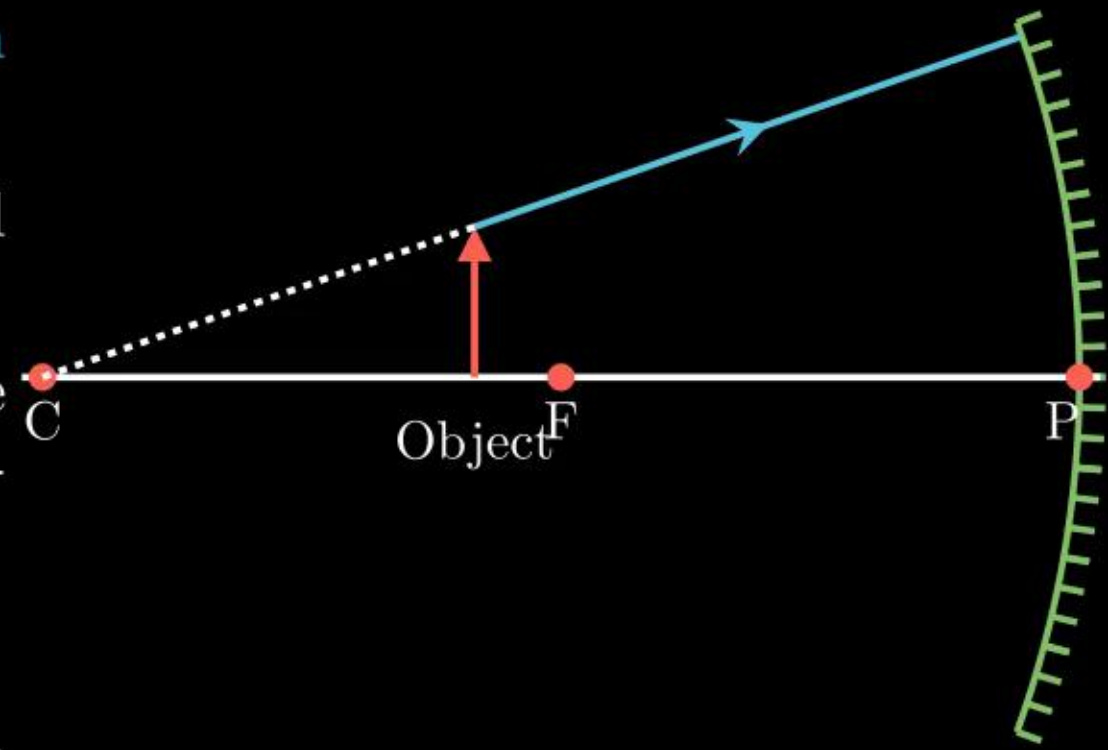
Example 8 : Magnification produced by a rear-view mirror fitted in vehicles:

- (i) is less than one
- (ii) is more than one
- (iii) is equal to one
- (iv) can be more than one or less than one depending upon the position of the object in front of it.

Solution:

Example 9 : While looking at the diagram Nalini concluded the following:

- (i) The image of the object will be a virtual one.
- (ii) The reflected ray will travel along the same path as the incident ray but in opposite direction.
- (iii) The image of the object will be inverted.
- (iv) This is a concave mirror and hence the focal length will be negative.



Which one of the above statements are correct?

- | | |
|--------------------------|-------------------------------|
| (a) (i) and (ii) | (b) (i) and (iii) |
| (c) (ii), (iii) and (iv) | (d) (i), (ii), (iii) and (iv) |

Example 10 : A real image, $\frac{1}{5}$ th size of object is formed at a distance of 18 cm from a mirror. What is the nature of the mirror? Calculate its focal length?

Solution :

Example 11 : We wish to obtain an erect image of an object, using a concave mirror of focal length 15 cm. What should be the range of distance of the object from the mirror? What is the nature of the image? Is the image larger or smaller than the object? Draw a ray diagram to show the image formation in this case

Solution :

Example 12 : An object is placed at a distance of 10 cm from a convex mirror of focal length 15 cm. Find the position and nature of the image.

Solution :

Example 13 : The magnification produced by a plane mirror is $+1$. What does this mean?.

Solution :

Example 14 : An object 5.0 cm in length is placed at a distance of 20 cm in front of a convex mirror of radius of curvature 30 cm. Find the position of the image, its nature and size.

Solution :

Example 15 : An object of size 7.0 cm is placed at 27 cm in front of a concave mirror of focal length 18 cm. At what distance from the mirror should a screen be placed, so that a sharp focussed image can be obtained? Find the size and the nature of the image.

Solution :

Example 15 : An object is placed at a distance of 30 cm in front of a convex mirror of focal length 15 cm. Write characteristics of the image formed by the mirror. (Delhi 2017)

Solution :

Four characteristics of the image formed by the given convex mirror are :

- (i) Virtual.
- (ii) Erect
- (iii) Diminished
- (iv) Image is always formed behind the mirror between pole and focus.